

APACHE ICEBERG: THE MODERN TABLE FORMAT



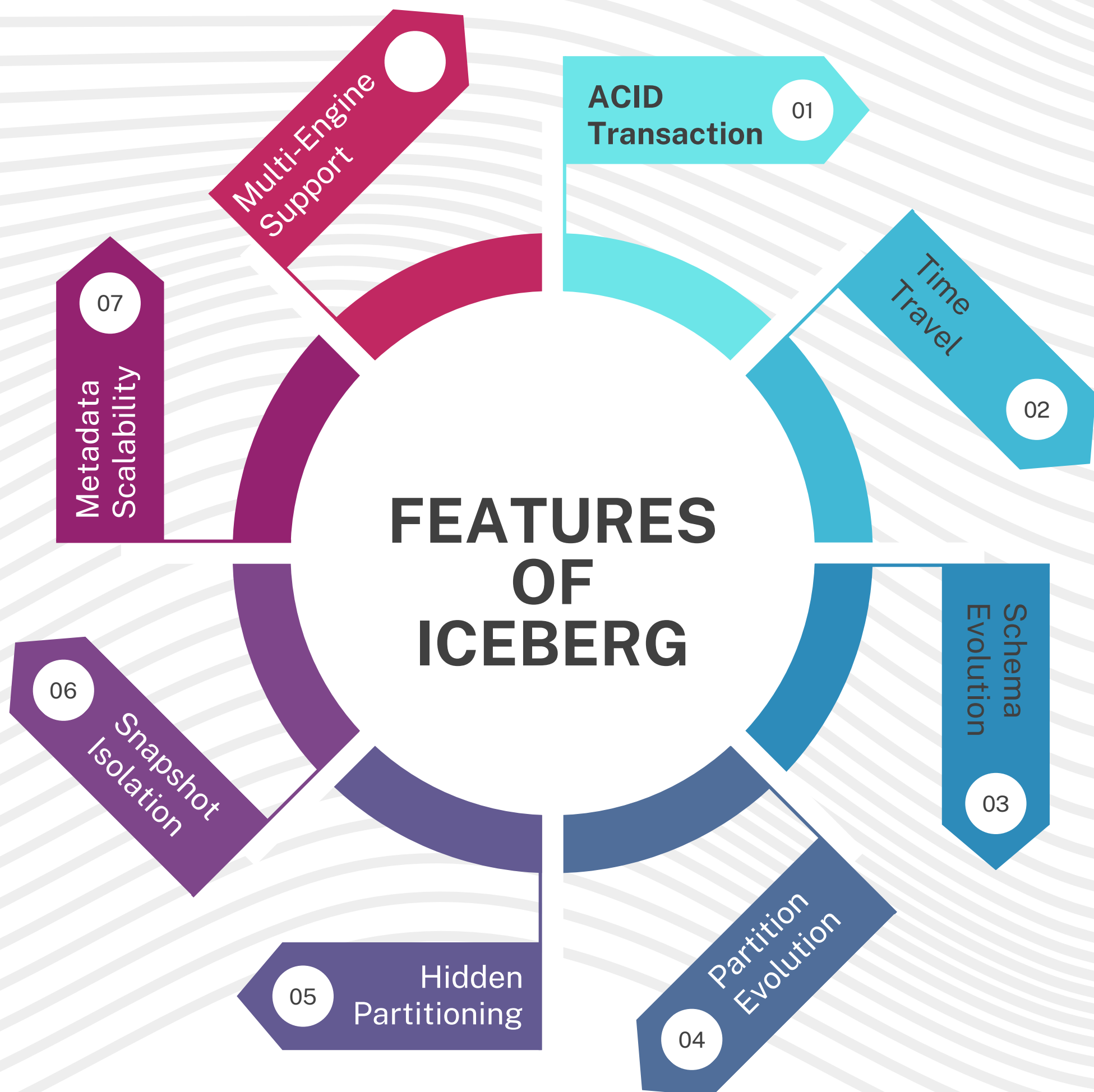
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WHAT IS APACHE ICEBERG?

Apache Iceberg is an open-source table format for data lakes that enables data stored in files (Parquet, ORC, Avro) to behave like a database table while maintaining the scalability and low cost of a data lake.

Think of it as:

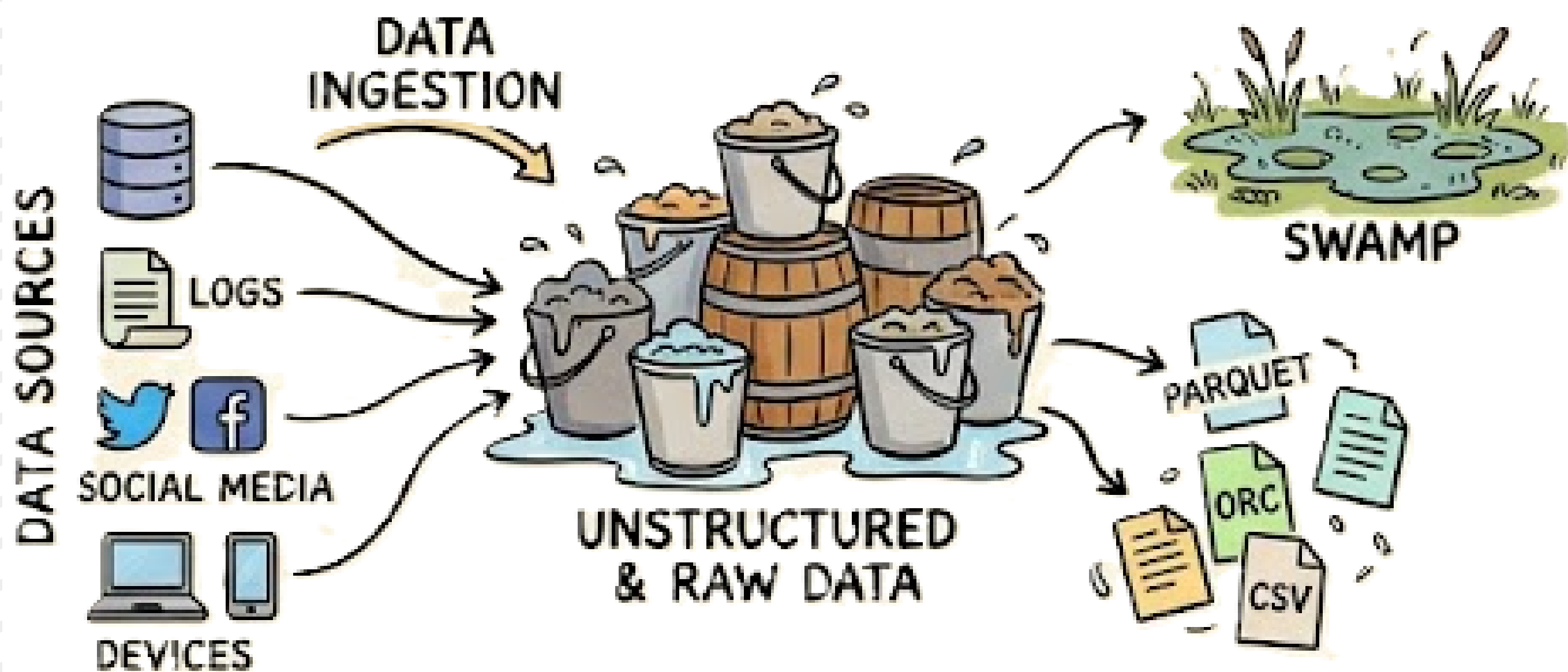
Data Warehouse Features + Data Lake Storage



WHY ICEBERG?

Problems with Traditional Data Lakes

- ✗ No transactions
- ✗ Slow metadata operations
- ✗ Difficult schema changes
- ✗ Partition management issues
- ✗ Concurrent write conflicts

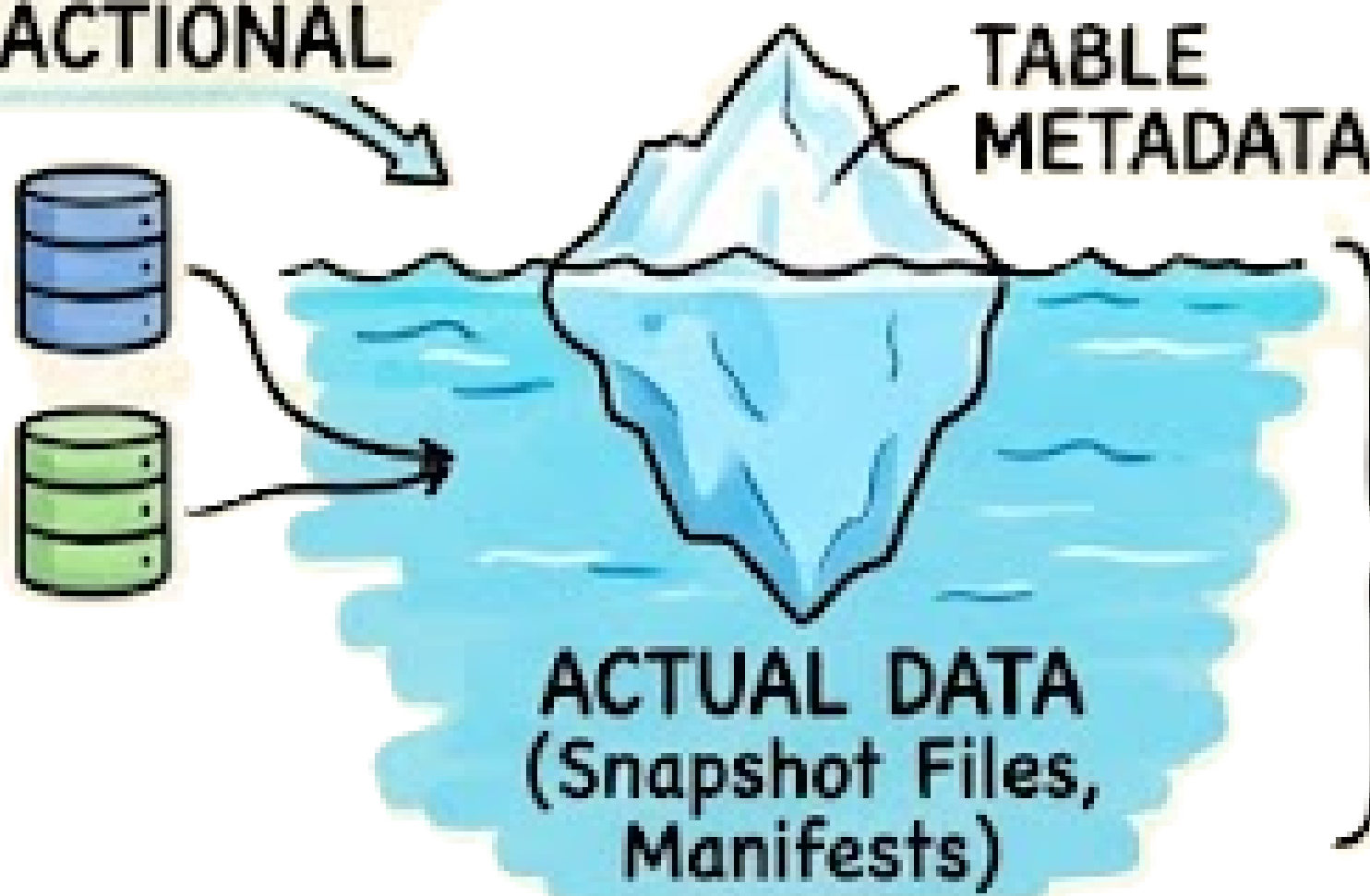


HOW ICEBERG IS BETTER THAN DATA LAKES?

Apache Iceberg is an open table format that adds a metadata layer on top of data lake files.

- It organizes data using metadata, snapshots, and manifest files for efficient management.
- Iceberg converts raw files into managed tables that behave like database tables.
- It provides ACID compliance, ensuring safe and reliable concurrent reads and writes.
- Its snapshot-based architecture enables Time Travel to access historical versions of data.
- As a result, Iceberg delivers better reliability, governance, and scalability for modern data lakes.

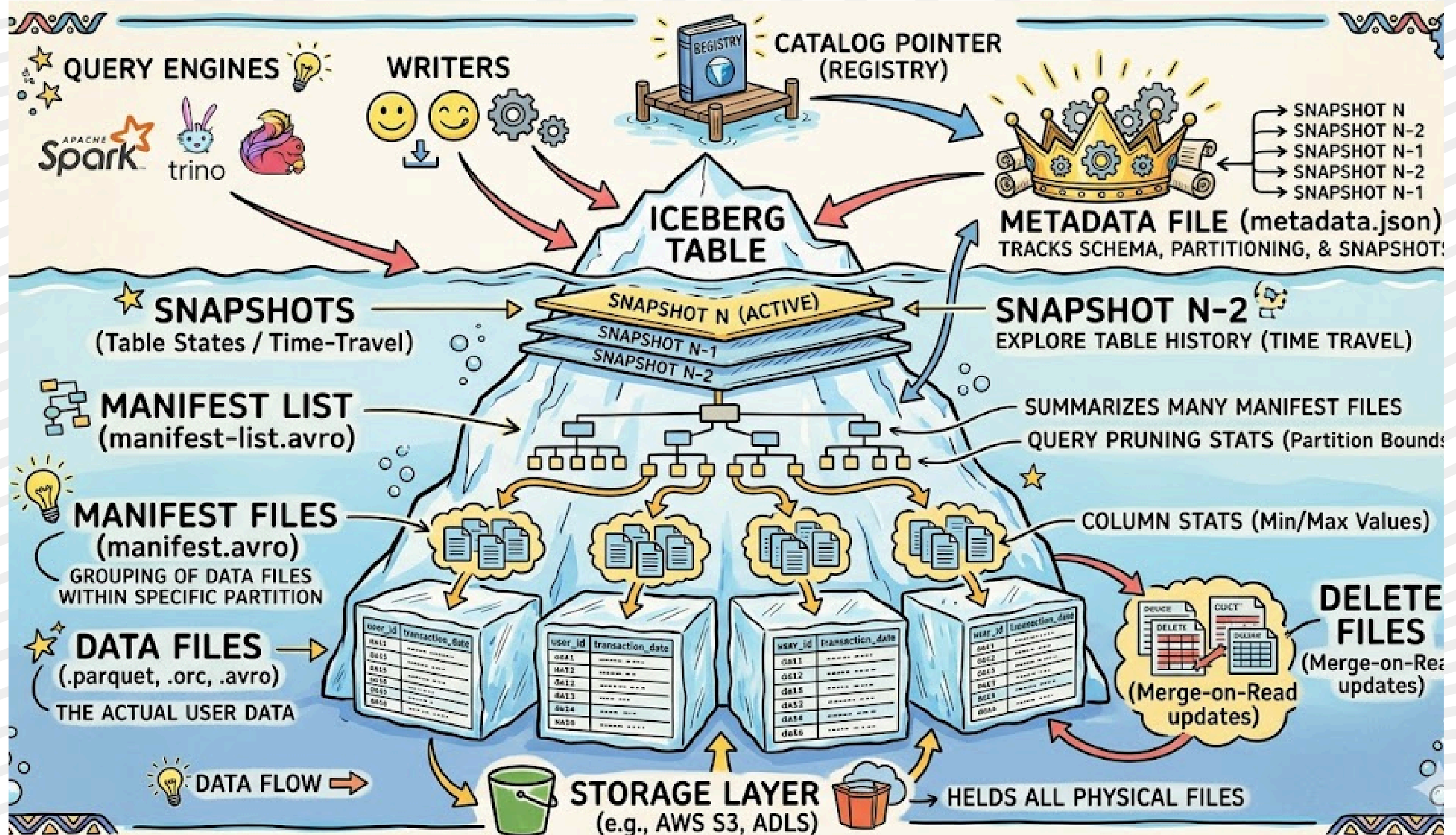
**STRUCTURED DATA,
TRANSACTIONAL**



ICEBERG ARCHITECTURE

Iceberg adds a structured metadata layer over standard data lake files.

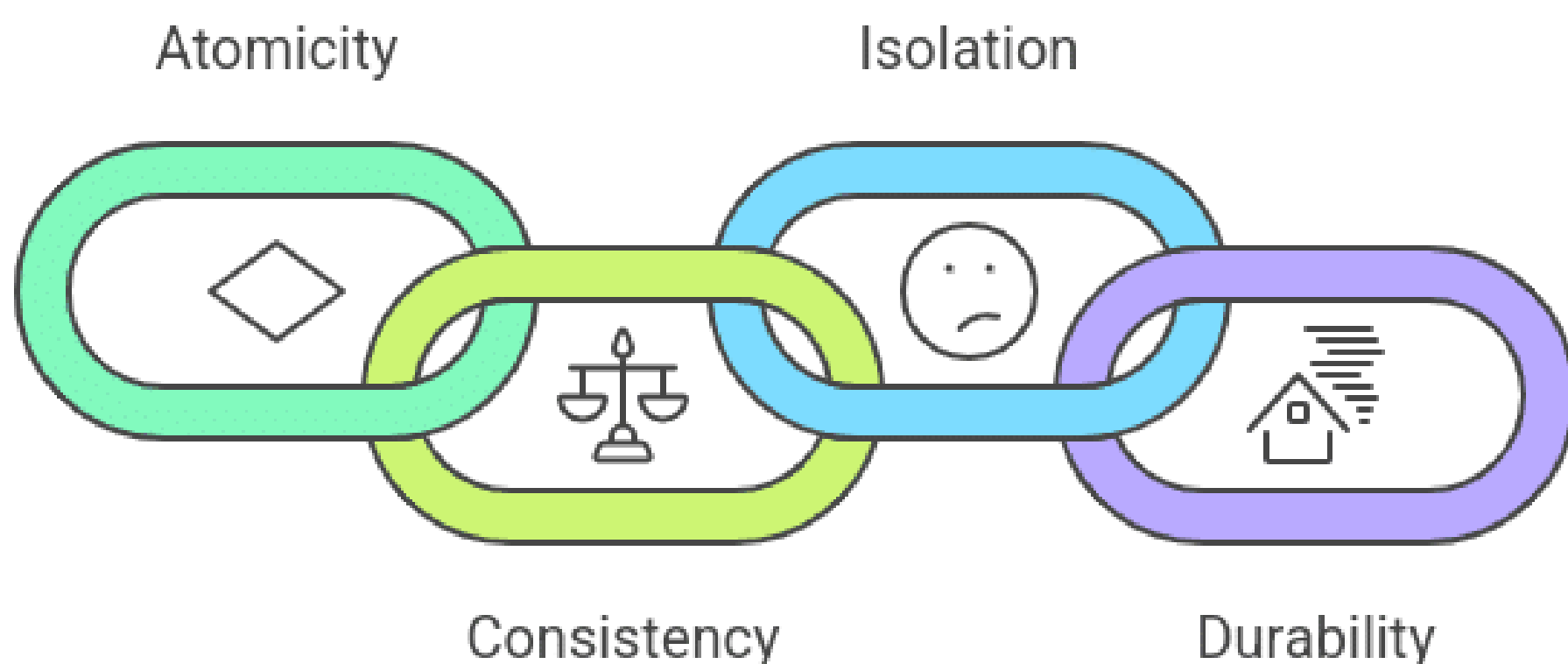
- A central catalog tracks the active metadata file for schema and snapshots.
- Snapshots use a manifest list to enable ACID transactions and time travel.
- This list indexes manifest files that manage specific data groups.
- Manifest files track exact file paths and statistics for faster queries.
- Finally, these route the query engine directly to the physical storage files.



ACID TRANSACTIONS

Multiple concurrent writers can attempt to modify an Iceberg table simultaneously. The goal is to ensure:

- Atomicity: Each commit is either fully completed or not applied at all. A failure during the write process should leave the table in its original state.
- Consistency: The table transitions only from one consistent state to another, upholding defined rules.
- Isolation: Concurrent transactions should seem as if they ran sequentially, isolated from each other.
- Durability: Once a commit succeeds, its changes must persist reliably, even after system failures.



SCHEMA EVOLUTION

Modify schema without rewriting data.

Supported Changes

- ✓ Add columns
- ✓ Rename columns
- ✓ Reorder columns
- ✓ Delete columns
- ✓ Type promotion



```
ALTER TABLE sales  
ADD COLUMN region STRING;
```

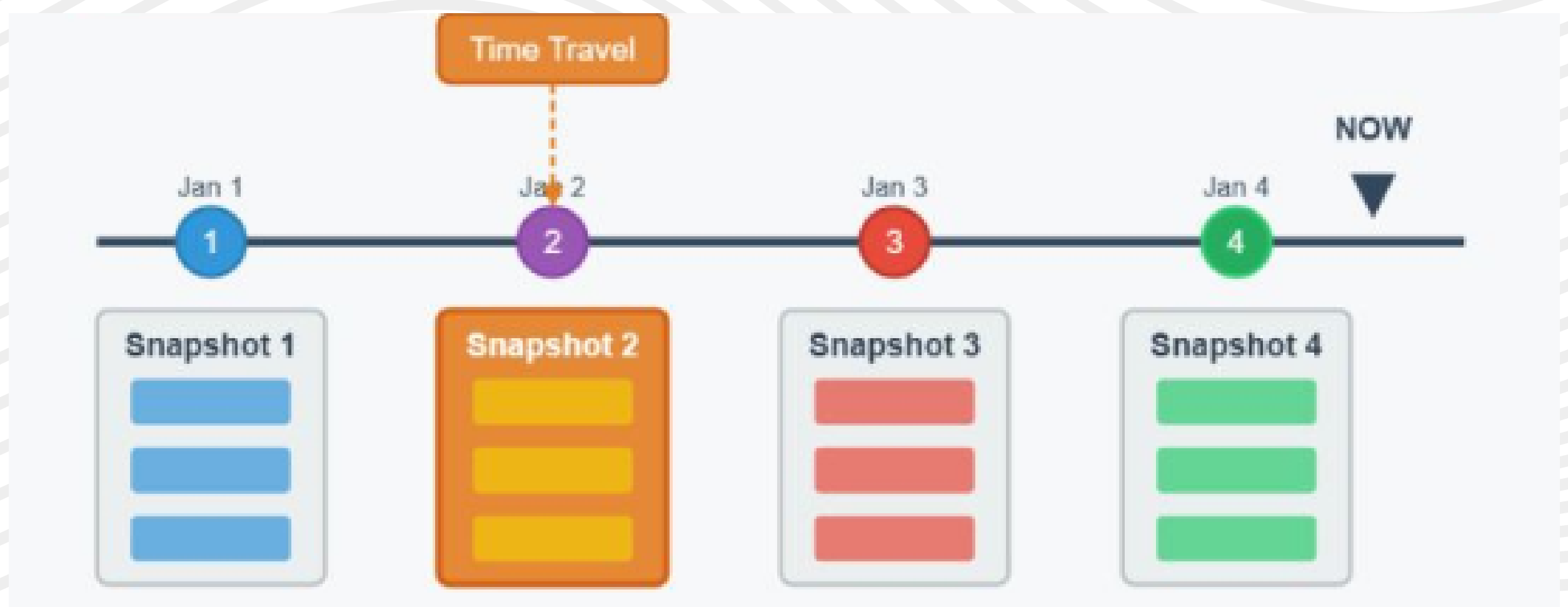
TIME TRAVEL

Query historical versions of data.

Time-travel enables reproducible queries that use exactly the same table snapshot, or lets users easily examine changes. Version rollback allows users to quickly correct problems by resetting tables to a good state.



```
SELECT * FROM my_table TIMESTAMP AS OF  
'2026-05-26 13:21:00';
```



ICEBERG ECOSYSTEM

